

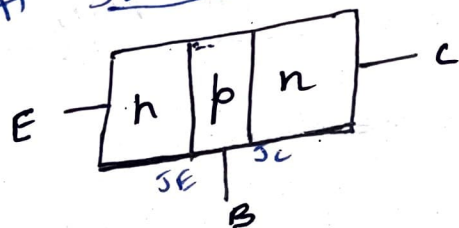
Transistor

→ Transistor is a three terminal electronic device.

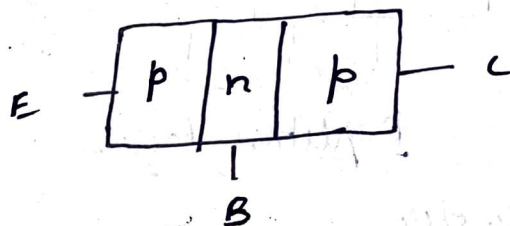
Bipolar Junction Transistor (BJT)

It is called a Bipolar Device because current flow due to both majority and minority carriers.

Structures of BJT



npn



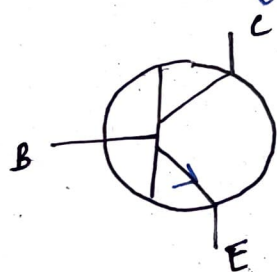
pnp

Emitter (E) - Heavily doped

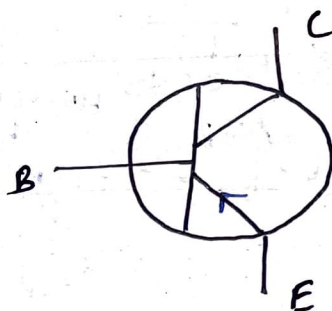
Collector (C) - Moderately doped [largest in size]

Base (B) - Lightly doped [narrow width]

Symbols of BJT



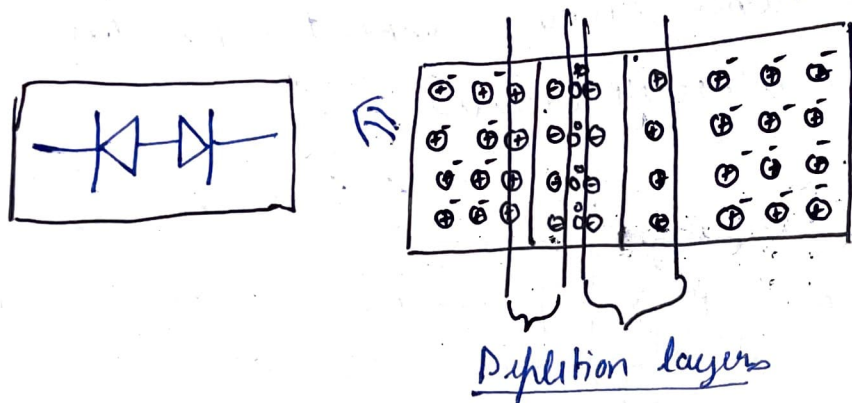
npn



pnp

Unbiased Transistor

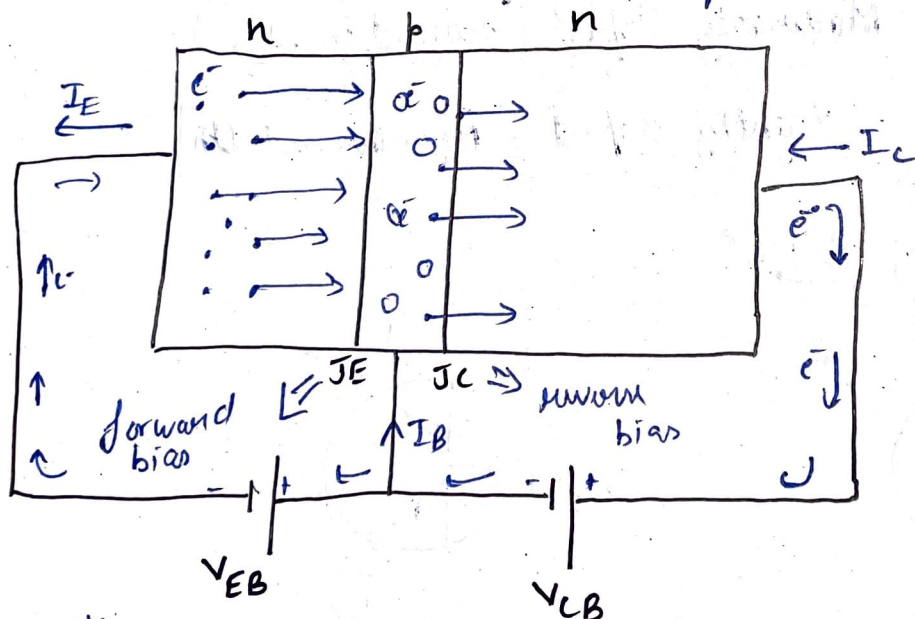
When no external supply is connected to a transistor the transistor is in unbiased condition.



2 depletion layers are formed due to recombination of e^- s and holes

Biased Transistor

External supply is connected to the transistor. We get currents from different parts of the transistor. Some e^- s recombine, some pass to C .



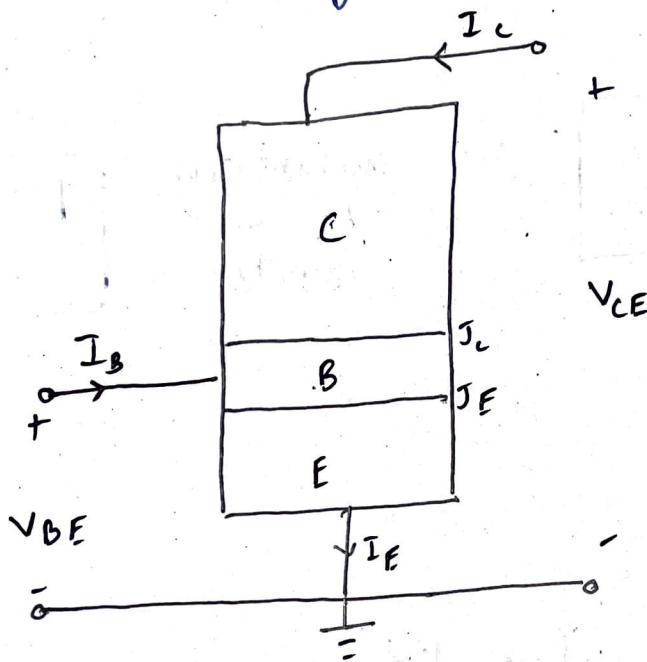
⇒ in emitter region, e^- was moving due to drift, so drift current is produced.

⇒ in base region, e^- was moving due to diffusion, so it is called as diffusion current.

⇒ in collector region, there is drift current

Configuration	I/P node	O/P node
1) Common Base	E	C
2) Common Emitter	B	C
3) Common Collector	B	E

Common Emitter Configuration



$$\begin{aligned} \text{I/P current} &= I_B \\ \text{I/P voltage} &= V_{BE} \end{aligned}$$

$$\begin{aligned} \text{O/P current} &= I_C \\ \text{O/P voltage} &= V_{CE} \end{aligned}$$

$$\Rightarrow I_C = f(I_B)$$

$$\text{We have from CBC} \Rightarrow I_C = \alpha I_E + I_{CBO} \quad (1)$$

$$I_E = I_B + I_C \quad (2)$$

using (1) & (2)

$$\Rightarrow I_C = \alpha (I_B + I_C) + I_{CBO}$$

$$\Rightarrow (1 - \alpha) I_C = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_C = \left(\frac{\alpha}{1 - \alpha} \right) I_B + \frac{I_{CBO}}{1 - \alpha}$$

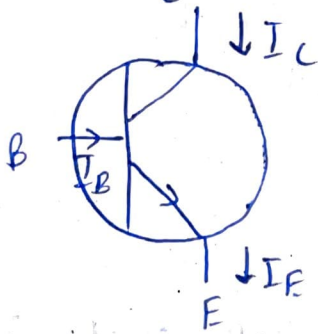
$$\Rightarrow I_C = (\beta) I_B + \underbrace{I_{CBO} (\beta + 1)}_{I_{CEO}}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta + 1 = \frac{1}{1 - \alpha}$$

$\Rightarrow$ imp. possibility of p -region is continuously decreasing due to recombination by e^- s from n -region. There will come a time when attraction stops and diffusion will stop and current will also stop, so we need to provide some holes which will be produced by current I_B .

$\Rightarrow$ imp. this way we can control I_C through I_B .



$$\Rightarrow I_E = I_B + I_C$$

current gain
for
common base

$$\alpha = \frac{I_C}{I_E}$$

current gain
for common
emitter

$$\beta = \frac{I_C}{I_B}$$

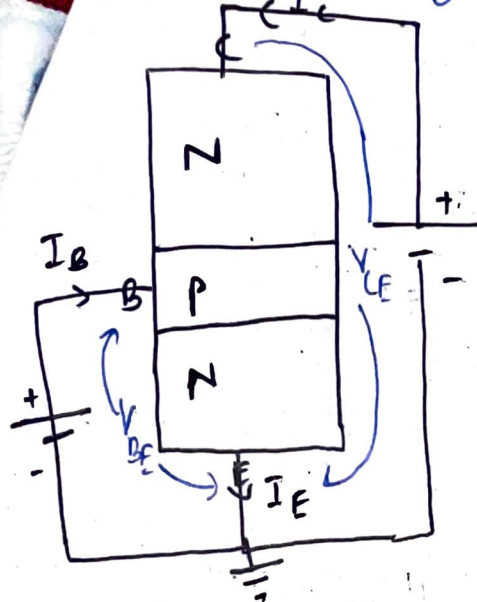
$$\Rightarrow I_E = I_C + I_B$$

$$\Rightarrow \frac{I_E}{I_C} = 1 + \frac{I_B}{I_C}$$

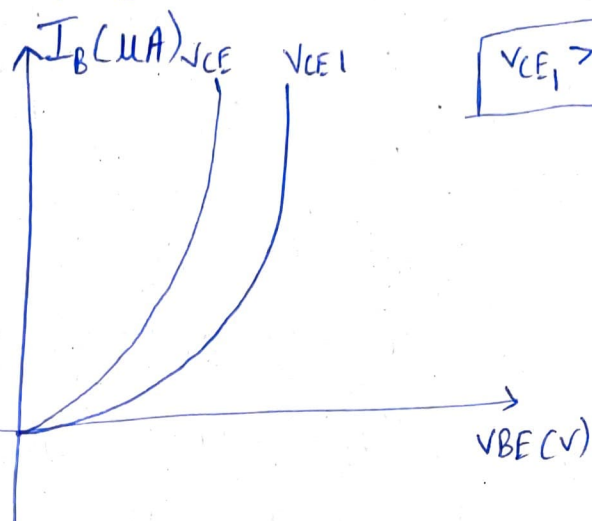
$$\Rightarrow \frac{1}{\alpha} = 1 + \frac{1}{\beta} \Rightarrow \frac{1}{\beta} = \frac{1-\alpha}{\alpha}$$

$$\Rightarrow \alpha = \frac{\beta}{1+\beta}$$

$$\Rightarrow \beta = \frac{\alpha}{1-\alpha}$$



1) Curve b/w I/P voltage & I/P current keeping o/p voltage constant

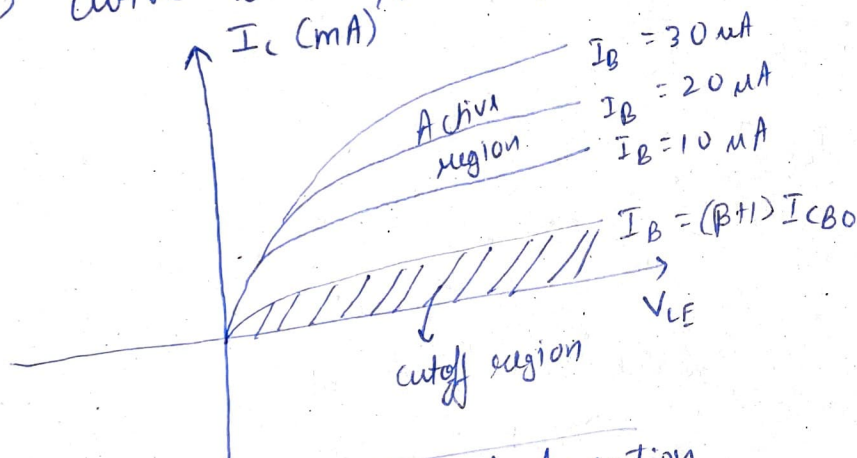


$$V_{CE1} > V_{CE2}$$

$V_{CE} \uparrow \Rightarrow I_{B, eff} \downarrow$

Output characteristics of CE

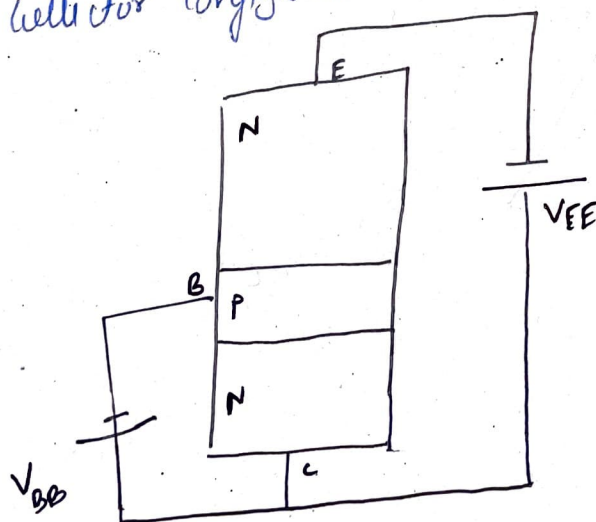
1) Curve b/w O/P voltage & O/P current with const. I/P current

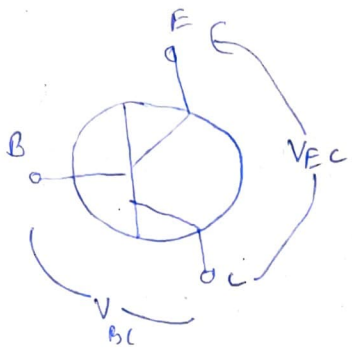


$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

$$\Rightarrow I_B = 0 \Rightarrow I_C = (\beta + 1) I_{CBO}$$

Common collector configuration





current amplification factor

$$\gamma = \frac{I_E}{I_B}$$

$$I_E = f(I_B)$$

$$\Rightarrow I_E = I_B + I_C$$

$$= I_B + \alpha I_E + I_{CBO}$$

$$\Rightarrow (1 - \alpha) I_E = I_B + I_{CBO}$$

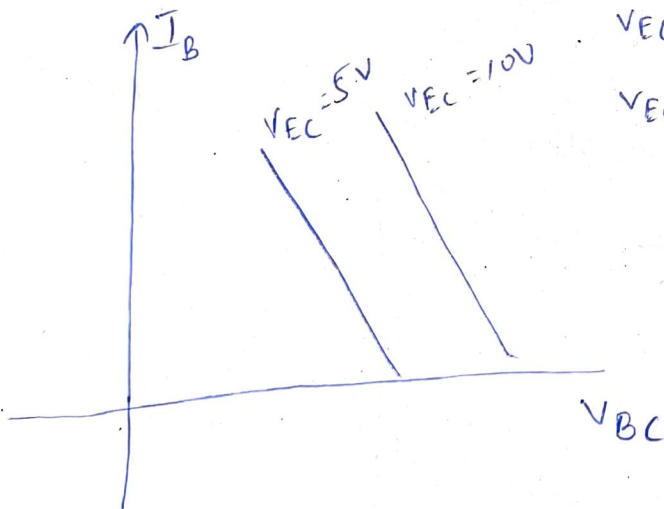
$$\Rightarrow I_E = \frac{1}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha} = (\beta + 1) [I_B + I_{CBO}]$$

$$\Rightarrow I_E = (\beta + 1) [I_B + I_{CBO}]$$

$$\Rightarrow \frac{I_E}{I_B} = (\beta + 1) \left[1 + \frac{I_{CBO}}{I_B} \right] \Rightarrow \boxed{\gamma = 1 + \beta}$$

$$\Rightarrow \boxed{I_E = \gamma [I_B + I_{CBO}]}$$

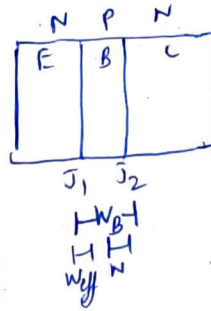
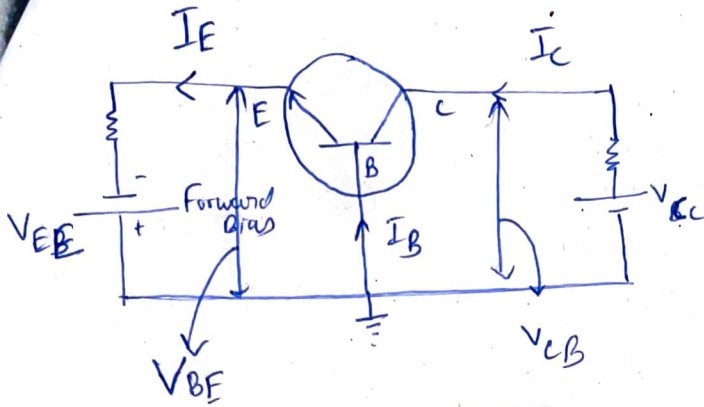
Input characteristics of CC



$$V_{EC} = V_{BC} + V_{BE}$$

$$V_{EC} \uparrow \Rightarrow I_B \downarrow$$

Common Base Configuration



$J_1 = FB, J_2 = RB$
 $V_{CB} \uparrow \Rightarrow R \cdot B \uparrow$
 $\downarrow$
 W increases
 $\Rightarrow W_B = W_{eff} + W$
 $\Rightarrow W_{eff} = W_B - W$

Base Modulation $\Rightarrow W_{eff} = \downarrow$

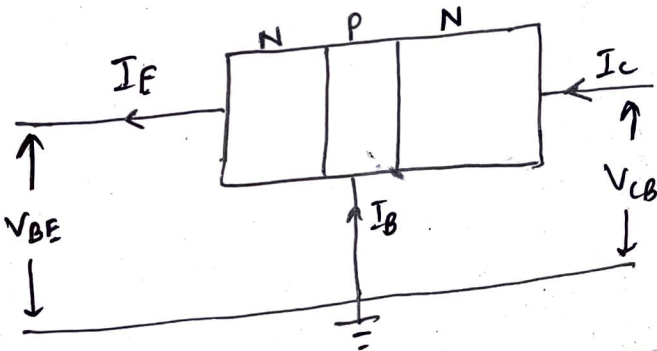
$W_{eff} \downarrow \Rightarrow$ less recombination
 $\Rightarrow$ more e^- pass through base
 $\Rightarrow \alpha \uparrow \Rightarrow I_C = \alpha I_E + I_{CBO}$
 $\Rightarrow I_C \uparrow, I_E \uparrow$

Early effect

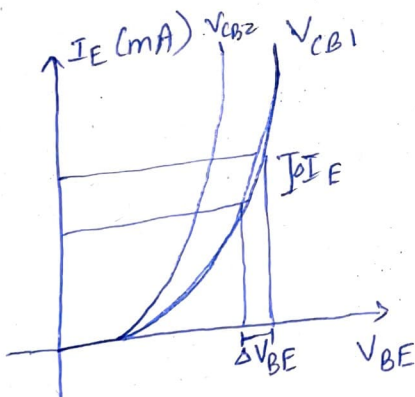
Characteristics for Common Base Configuration

Curves

1) Input voltage vs / Input current when O/P voltage is const.



I/P voltage = V_{BE}
 I/P current = I_E
 O/P voltage = V_{CB}
 O/P current = I_C



here $V_{CB2} > V_{CB1}$

Dynamic input resistance

$$= \frac{\text{change in I/P voltage}}{\text{change in I/P current}} = \frac{\Delta V_{BE}}{\Delta I_E}$$

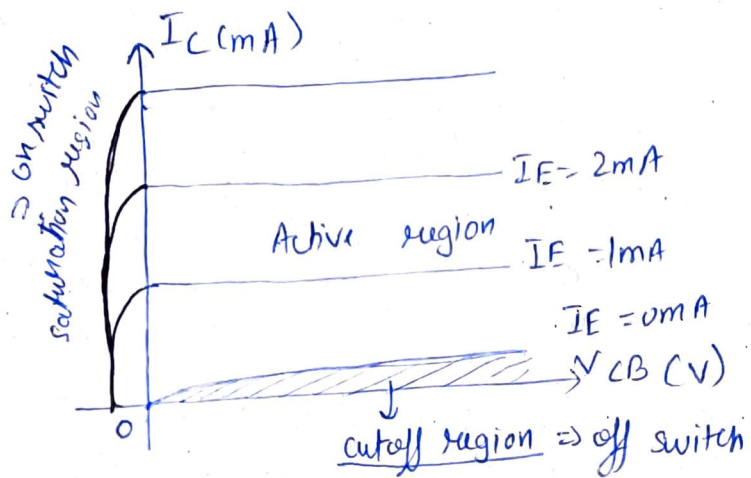
2) Output Characteristics

O/P voltage

V_{CB}

O/P current

keeping I/P current constant



$$I_C = \alpha I_E + I_{CBO}$$

$\Rightarrow$ wenn $I_E = 0$

$$I_C = -I_{CBO}$$

$$I_C = \alpha I_E + I_{CBO}$$

$$I_E \gg I_{CBO}$$

$$I_C \approx I_E$$

Relation between α , β & γ

$$\alpha = \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B}, \quad \gamma = \frac{I_E}{I_B}$$

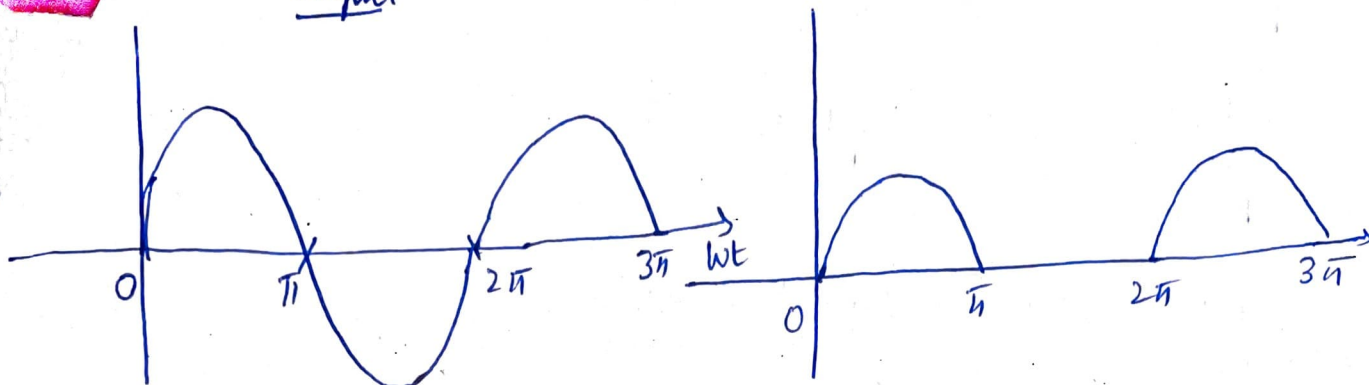
$$I_E = I_C + I_B$$

$$\Rightarrow \frac{I_E}{I_B} = \frac{I_C}{I_B} + 1 \Rightarrow \boxed{\gamma = \beta + 1}$$

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

$$\Rightarrow \gamma = \frac{\alpha}{1-\alpha} + 1 = \frac{1}{1-\alpha} = \gamma$$

$$\Rightarrow \boxed{\gamma = \beta + 1 = \frac{1}{1-\alpha}} \Rightarrow \text{relation}$$

Input

$$v_i = v_m \sin \omega t$$

Avg. voltage / DC

$$V_{avg} = \frac{\int_0^{2\pi} v_o d\omega t}{2\pi} = \int_0^{\pi} v_o d\omega t + \int_{\pi}^{2\pi} v_o d\omega t$$

$$= \int_0^{\pi} v_m \sin \omega t \cdot d\omega t + \int_{\pi}^{2\pi} v_m \sin \omega t \cdot d\omega t$$

$$V_{avg} = \frac{v_m}{\pi}$$

$$\Rightarrow I_{avg} = \frac{I_m}{\pi}$$

$$V_{rms} = \sqrt{\frac{\int_0^{2\pi} v_o^2 d\omega t}{2\pi}} = \sqrt{\frac{\int_0^{2\pi} v_m^2 \sin^2 \omega t \cdot d\omega t}{2\pi}}$$

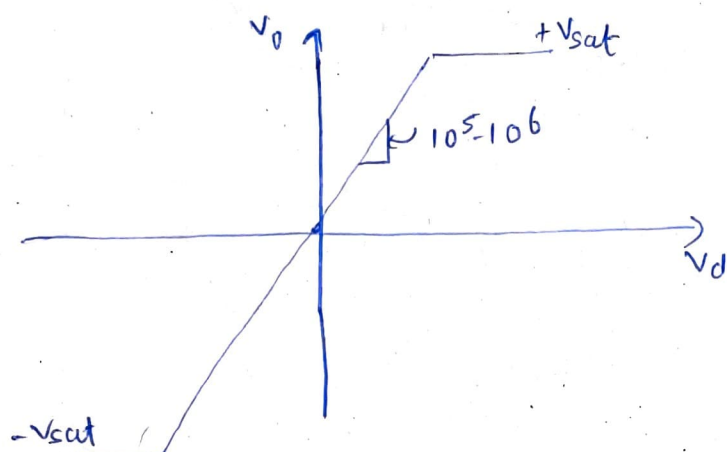
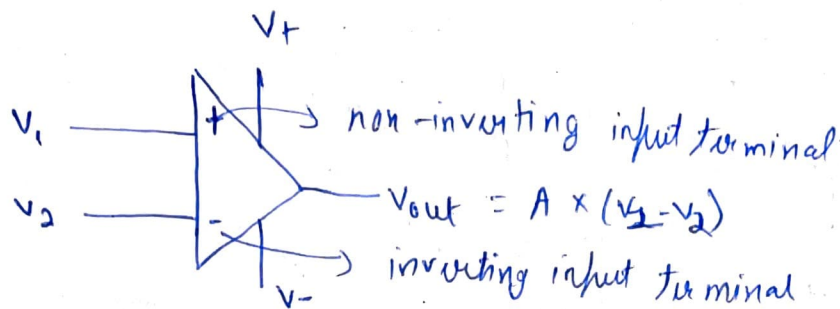
$$= \frac{v_m}{\sqrt{2\pi}} \sqrt{\int_0^{2\pi} \sin^2 \omega t \cdot d\omega t}$$

$$= \frac{v_m}{\sqrt{2\pi}} \sqrt{\int_0^{\pi} \sin^2 \omega t \cdot d\omega t + \int_{\pi}^{2\pi} 0 \cdot d\omega t} = \frac{v_m}{\sqrt{2\pi}} \sqrt{\int_0^{\pi} \frac{1 - \cos 2\omega t}{2} \cdot d\omega t}$$

$$= \frac{v_m}{\sqrt{2\pi}} \sqrt{\frac{\pi}{2} - \int_0^{\pi} \frac{\cos 2\omega t \cdot d\omega t}{2}} = \frac{v_m}{\sqrt{2\pi}} \sqrt{\frac{\pi}{2} - \frac{1}{2} [\sin 2\omega t]_0^{\pi}}$$

$$= \sqrt{\frac{v_m^2}{2}}$$

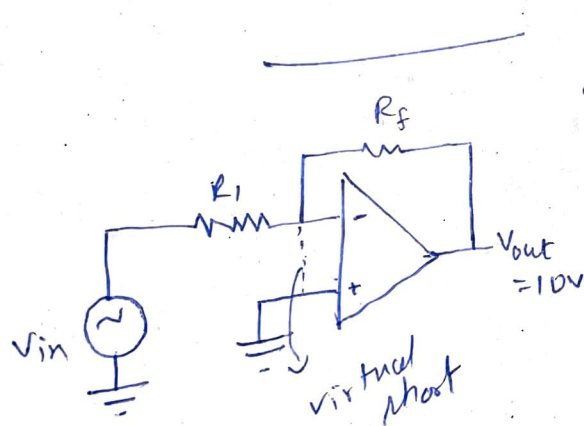
$$\Rightarrow V_{rms} = \frac{v_m}{2}$$



$$V_{sat} < V_f$$

ideal op-amp

- 1) $R_i = \infty$, 2) $R_o = 0$, 3) $A = \infty$, 4) $V_{out} = 0$ when $V_{in} = 0$
- 5) $B = \infty$, 6) slew rate $= \infty$, 7) CMRR $= \infty$



Inverting op-amp config

$$A = 10^6$$

$$V_{out} = A \times V_d$$

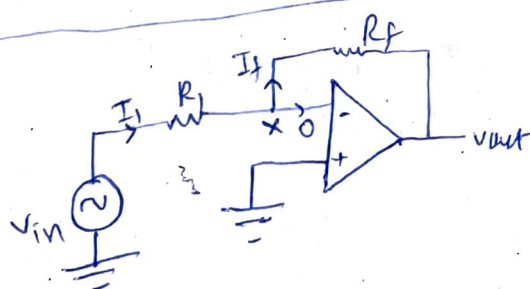
$$10 = 10^6 V_d$$

$$\Rightarrow V_d = 10 \mu V$$

$$V^+ - V^- = 10 \mu V$$

$$\approx 0$$

$$\Rightarrow V^+ = V^-$$



$$R_{in} = \infty$$

$$\Rightarrow I_1 = I_f$$

$$\Rightarrow \frac{V_{in} - V_x}{R_1} = \frac{V_x - V_{out}}{R_f}$$

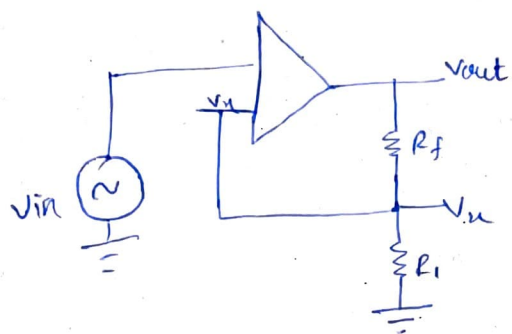
$$\Rightarrow \frac{V_{in}}{R_1} = -\frac{V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{in}}{V_{out}} = -\frac{R_1}{R_f} \Rightarrow \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$$

since $V_x = 0$ [grounded]

closed loop gain

Non-inverting Operational Amplifier

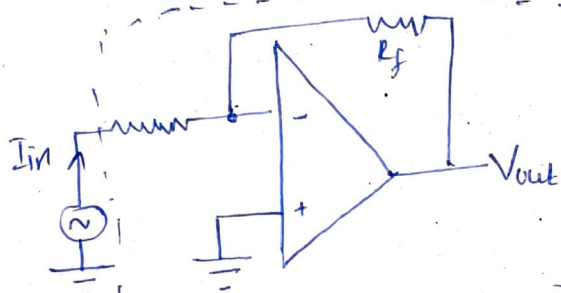


$$V_x = \frac{R_i}{R_i + R_f} \times V_{out}$$

$$V_x \approx V_{in}$$

$$\Rightarrow \frac{R_i}{R_i + R_f} \times V_{out} = V_{in} \Rightarrow \boxed{\frac{V_{out}}{V_{in}} = \left(1 + \frac{R_f}{R_i}\right)}$$

Input Impedance of inverting op-amp

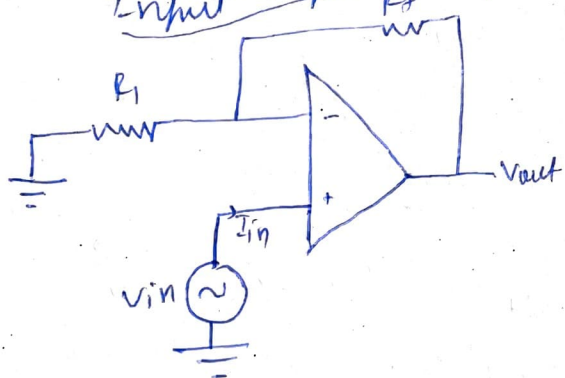


$$Z_{in} = \frac{V_{in}}{I_{in}}$$

$$\Rightarrow I_{in} = \frac{V_{in}}{R_i}$$

$$\Rightarrow \boxed{R_i = \frac{V_{in}}{I_{in}}}$$

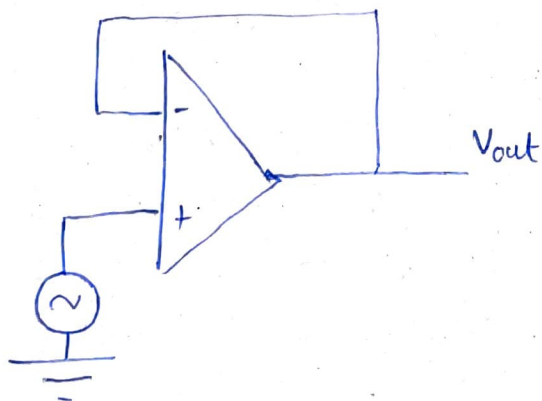
Input impedance of non-inverting op-amp



$$I_{in} = 0$$

$$\boxed{Z_{in} = \infty}$$

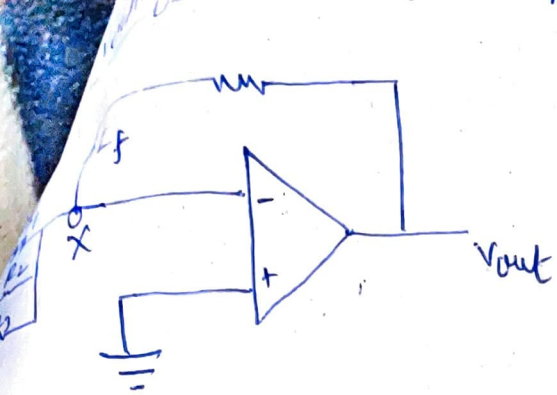
op-amp as buffer



$$R_f = 0$$

$$R_i = \infty$$

$$\Rightarrow \boxed{V_{in} = V_{out}}$$



Applying KCL

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{0 - V_{out}}{R_f}$$

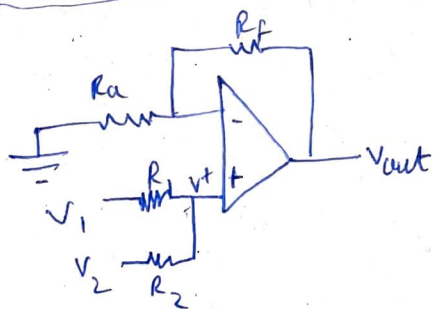
$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left[\frac{R_f V_1}{R_1} + \frac{R_f V_2}{R_2} + \frac{R_f V_3}{R_3} \right]$$

$$= - [A V_1 + B V_2 + C V_3]$$

$\swarrow$
 $\left(\frac{R_f}{R_i} \right)$

Non-Inverting Summing Amplifier



$$V_{out} = \left(1 + \frac{R_f}{R_a} \right) V^+$$

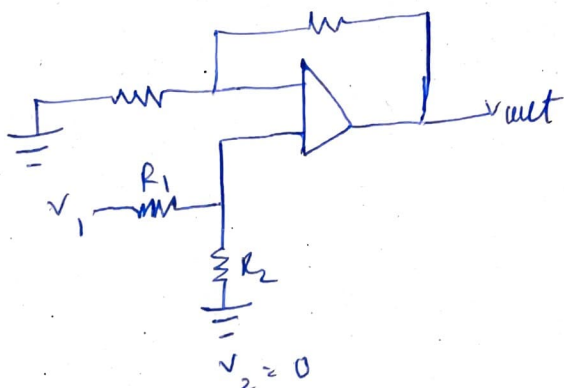
using superposition

$$V_1^+ = \frac{R_2 V_1}{R_1 + R_2}$$

similarly

$$V_2^+ = \frac{R_1 V_2}{R_1 + R_2}$$

$$\Rightarrow V^+ = V_1^+ + V_2^+ = \frac{R_1 V_2 + R_2 V_1}{R_1 + R_2}$$



$$\Rightarrow V_{out} = \left(1 + \frac{R_f}{R_a} \right) \left(\frac{R_2 V_1 + R_1 V_2}{R_1 + R_2} \right)$$

$$R_1 = R_2 = R$$

$$V_{out} = \left(1 + \frac{R_f}{R_a} \right) \left(\frac{V_1 + V_2}{2} \right)$$

$$R_f = R_a$$

$$\Rightarrow \boxed{V_{out} = V_1 + V_2}$$